

ChurnReaper

Explainable AI-Powered Customer Churn Prediction System

Project Report

Submitted by

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1. Executive Summary

ChurnReaper is an end-to-end Explainable AI system designed to predict customer churn in the telecom industry. The system integrates machine learning, SHAP-based explainability, and an interactive Dash dashboard to deliver actionable business insights. Built on the IBM Telco Customer Churn dataset with 7,043 customers and 21 features, ChurnReaper achieves over 80% prediction accuracy and identifies the key drivers behind customer attrition.

Metric	Value
Dataset size	7,043 customers, 21 features
Churn rate	~26.5% (1,869 customers)
Model accuracy (Random Forest)	~80.4%
Model accuracy (Logistic Regression)	~80.6%
ROC-AUC score	~84.3%
Top churn driver	Total Charges
Dashboard technology	Plotly Dash
Explainability method	SHAP (SHapley Additive exPlanations)

2. Project Overview

2.1 Problem Statement

Customer churn — the rate at which customers stop doing business with a company — is one of the most critical challenges for subscription-based businesses. In the telecom sector, acquiring a new customer costs five to ten times more than retaining an existing one. Despite this, many organizations lack the tools to predict churn before it happens or to understand why customers leave.

2.2 Objective

ChurnReaper addresses this problem by providing a complete, production-ready system that:

- Predicts whether a customer is at risk of churning
- Explains the specific reasons behind each prediction using SHAP
- Presents all findings through an interactive business dashboard
- Enables real-time churn probability scoring for individual customers

2.3 System Architecture

The project follows a modular architecture with clear separation of concerns:

Module	File	Responsibility
Data preprocessing	src/preprocessing.py	Cleaning, encoding, feature engineering
Model training	src/train.py	Training and hyperparameter tuning
Random Forest	src/random_forest.py	Random Forest model implementation
Feature importance	src/feature_importance.py	Extracting and ranking features
Explainability	src/explain.py	SHAP value computation
Prediction engine	src/predict.py	Real-time inference pipeline
Dashboard	dashboard/app.py	Interactive Dash web application

3. Dataset

3.1 Dataset Overview

The project uses the IBM Telco Customer Churn dataset, publicly available on Kaggle. It represents a fictional but realistic telecom company's customer base.

Attribute	Detail
Source	Kaggle — IBM Telco Customer Churn
Total records	7,043 customers
Total features	21 original features
Target variable	Churn (Yes / No — binary classification)
Class distribution	~26.5% churned, ~73.5% retained
File format	CSV (WA_Fn-UseC_-Telco-Customer-Churn.csv)

3.2 Feature Categories

Customer Demographics

- CustomerID — unique identifier
- Gender — Male / Female
- SeniorCitizen — whether the customer is a senior citizen
- Partner — whether the customer has a partner
- Dependents — whether the customer has dependents

Service Information

- PhoneService, MultipleLines
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- InternetService — DSL, Fiber Optic, or None
- OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport
- StreamingTV, StreamingMovies

Account Information

- Contract — Month-to-Month, One Year, Two Year
- PaperlessBilling — Yes / No
- PaymentMethod — Electronic Check, Mailed Check, Bank Transfer, Credit Card
- MonthlyCharges — monthly amount charged
- TotalCharges — total amount charged to date
- Tenure — number of months the customer has been with the company

4. Exploratory Data Analysis

4.1 Structural EDA

Initial structural analysis confirmed the dataset was largely clean with only 11 missing values in the TotalCharges column. These were customers with zero tenure (new customers) and were removed from the dataset.

Check	Finding
Shape	7,043 rows x 21 columns
Missing values	11 in TotalCharges (0.16%) — removed
Duplicate rows	0 duplicates found
Data types	TotalCharges stored as object — converted to float
Target balance	Imbalanced — 26.5% churn vs 73.5% retained

4.2 Analytical EDA — Key Findings

Churn by Contract Type

Month-to-month contract customers showed the highest churn rate at approximately 42%, compared to 11% for one-year and just 3% for two-year contracts. This was the strongest categorical predictor of churn.

Tenure vs Churn

Customers with shorter tenure (0 to 12 months) were significantly more likely to churn. The median tenure for churned customers was 10 months compared to 38 months for retained customers.

Monthly Charges vs Churn

Higher monthly charges correlated strongly with churn. Churned customers had a mean monthly charge of \$74.44 compared to \$61.27 for retained customers.

Payment Method vs Churn

Customers paying by electronic check had the highest churn at 45%, nearly double the rate of customers using bank transfer or credit card automatic payments.

Internet Service vs Churn

Fiber optic internet customers churned at approximately 41%, compared to 19% for DSL customers and 7% for customers with no internet service.

5. Machine Learning Models

5.1 Preprocessing Pipeline

- Removed 11 rows with missing TotalCharges
- Converted TotalCharges from string to float
- Label encoded binary categorical columns (Yes/No features)
- One-hot encoded multi-class categorical columns (Contract, PaymentMethod, InternetService)
- Scaled numerical features using StandardScaler
- Split dataset into 80% training and 20% test sets with stratification

5.2 Models Trained

Model	Accuracy	ROC-AUC	Notes
Logistic Regression	~80.6%	~84.3%	Fast, interpretable baseline
Random Forest	~80.4%	~84.3%	Selected for SHAP explainability

5.3 Model Selection Rationale

Random Forest was selected as the final model despite marginally lower accuracy because it provides native feature importance scores and integrates seamlessly with SHAP for per-prediction explanation. This aligns with the core business objective of explainability, not just prediction.

6. Explainable AI with SHAP

6.1 What is SHAP?

SHAP (SHapley Additive exPlanations) is a game-theory-based method for explaining machine learning model predictions. It assigns each feature a contribution score (SHAP value) for each individual prediction, allowing both global feature importance analysis and local per-customer explanations.

6.2 Top Churn Drivers Identified

Rank	Feature	Impact	Business Interpretation
1	TotalCharges	High	Long-term billing pressure drives attrition
2	Tenure	High	New customers are significantly more at risk
3	MonthlyCharges	High	High monthly bills correlate with churn
4	Electronic Check payment	Medium	Less committed payment method signals risk
5	Fiber Optic internet	Medium	Possibly dissatisfied with service quality
6	Contract type	Medium	Month-to-month = no commitment, easy to leave

6.3 Business Value of Explainability

The SHAP integration transforms ChurnReaper from a black-box predictor into a transparent decision-support tool. Business teams can understand exactly why the model flags a specific customer as at-risk, enabling targeted interventions such as offering a discounted long-term contract to a month-to-month fiber optic customer with high monthly charges.

7. Interactive Dashboard

7.1 Technology

The dashboard is built using Plotly Dash, a Python framework for building analytical web applications. It runs locally on `http://127.0.0.1:8051/` and requires no front-end knowledge to extend.

7.2 Dashboard Sections

Section	Content
Business metrics	Total customers, churned count, churn rate, revenue at risk
Churn distribution	Donut chart showing retained vs churned split

Section	Content
Monthly charges vs churn	Box plot comparing charge distributions
Feature importance	Bar chart of top predictive features
SHAP explainability	SHAP summary plot for global model interpretation
Prediction engine	Input form for real-time churn probability scoring

7.3 Prediction Engine

Users can enter a customer profile including tenure, monthly charges, contract type, internet service, and payment method and receive an instant churn prediction with probability score. This enables customer service teams to proactively identify and intervene with at-risk customers.

8. Project Structure

Folder / File	Purpose
dashboard/app.py	Main Dash application
dashboard/assets/	CSS and static assets
data/	Raw CSV dataset
models/churn_model.pkl	Saved trained model
models/random_forest.pkl	Saved Random Forest model
notebooks/	Jupyter notebooks for exploration
screenshots/	Dashboard screenshots for documentation
src/preprocessing.py	Data cleaning and feature engineering
src/train.py	Model training pipeline
src/random_forest.py	Random Forest implementation
src/feature_importance.py	Feature importance extraction
src/explain.py	SHAP explainability computation
src/predict.py	Real-time prediction pipeline
.gitignore	Excludes venv, pycache, sensitive files
Dockerfile	Container configuration for deployment
requirements.txt	All Python dependencies
README.md	Full project documentation

9. Technology Stack

Category	Technology	Purpose
Language	Python 3	Core programming language
Data processing	Pandas, NumPy	Data manipulation and analysis
Visualization	Matplotlib, Seaborn, Plotly	Charts and graphs
Machine learning	Scikit-learn	Model training and evaluation
Advanced ML	Random Forest, Logistic Regression	Classification models
Explainable AI	SHAP	Model interpretation and explanation
Dashboard	Plotly Dash	Interactive web application
Containerization	Docker	Deployment and portability
Version control	Git / GitHub	Code management and collaboration

10. Business Impact

ChurnReaper delivers measurable value across multiple dimensions of a telecom organization:

Business Goal	How ChurnReaper Addresses It
Identify at-risk customers	Real-time prediction engine scores individual customers
Understand churn drivers	SHAP explainability reveals root causes per customer
Improve retention strategies	Feature importance guides targeted intervention campaigns
Reduce revenue loss	Early warning allows proactive outreach before customers leave
Data-driven decisions	Dashboard provides executive-level visibility into churn metrics
Scalability	Dockerized deployment enables cloud hosting for team-wide access

11. Conclusion

ChurnReaper demonstrates how machine learning and explainable AI can be combined into a practical, business-ready tool. Rather than delivering a prediction in isolation, the system provides the full picture — who is at risk, how likely they are to leave, and most importantly why — giving organizations the information they need to act.

The project showcases a complete data science workflow from raw data exploration through model training, explainability analysis, and interactive dashboard deployment. The inclusion of a

Dockerfile and modular source code structure makes ChurnReaper ready for production deployment and extension.

Future enhancements could include deployment to a cloud platform such as Render or AWS, integration with a live CRM data feed, addition of a customer segmentation module using clustering, and automated weekly churn report generation.

GitHub: github.com/codewithshreyak-prog/ChurnReaper
